

NITHYA SAI DODLA

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PROFILE

Motivated B.Tech student in Artificial Intelligence with a strong foundation in programming, problem-solving, and software development. Experienced in building applications across multiple domains — from system-level C programming and hardware integration to data-driven Python projects. Comfortable working independently and in teams, managing tasks from design through implementation and testing. Quick learner with good communication skills and a genuine interest in technology, engineering, and delivering practical solutions.

EDUCATION

B.Tech, Artificial Intelligence — Amrita Vishwa Vidyapeetham

2023 – 2027 | Coimbatore, India

CGPA: 7.67 / 10

Senior Secondary — Sri Chaitanya Junior College

2021 – 2023 | Hyderabad, India

Marks: 96.8% | Mathematics, Physics, Chemistry

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Assembly, HTML, CSS

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Networking, Database Management, Hardware-Software Integration

Libraries & Frameworks: TensorFlow, PyTorch, NumPy, Pandas

Tools: Git, GitHub, VS Code, Jupyter Notebook, Arduino IDE, REST APIs

PROJECTS

Dual-Axis Solar Panel Tracking System

- Built an embedded system in C++ on Arduino Mega that automatically rotates a solar panel along two axes to track sunlight using real-time input from Light Dependent Resistors (LDRs) and servo motors.
- Handled full hardware-software integration: wiring sensor circuitry, writing motor control logic, calibrating tracking accuracy, and testing performance across different lighting conditions.
- Demonstrated practical application of electronics and programming — system consistently improved solar energy capture compared to a fixed panel.

Canteen Management System

- Developed a menu-driven C application to manage canteen operations including item listing, order placement, and automated bill generation — structured using modular functions with clear separation of responsibilities.
- Applied file handling, loops, and conditional logic to manage persistent data; tested all modules against edge cases to ensure reliability and correct output.

Proxy Server with Web Caching

- Implemented a proxy server using socket programming that intercepts HTTP requests, caches responses locally, and serves repeat requests from cache — reducing latency and network load.
- Built and tested client-server communication, request parsing, and cache management modules; debugged behaviour across different network conditions.

Friend Recommendation System

- Designed a recommendation engine using graph theory and adjacency matrices to suggest friends based on mutual connections — implemented in Python with clear, documented logic.
- Analysed algorithm performance on networks of varying sizes and explored optimisation strategies, strengthening understanding of data structures and computational complexity.

RELEVANT COURSEWORK

Data Structures & Algorithms | Object-Oriented Programming | Computer Networks | Operating Systems | Database Management Systems | Computer Architecture | Discrete Mathematics | Cloud Computing | Machine Learning | Computer Vision | Linear Algebra & Probability

SKILLS & ATTRIBUTES

- Strong programming fundamentals across multiple languages — C, C++, Python, Java — with hands-on project experience in each.
- Effective communicator and collaborator: comfortable in team environments, open to feedback, and experienced in contributing to shared codebases via Git/GitHub.
- Detail-oriented and systematic: approaches problems methodically — breaking down requirements, designing a solution, implementing, and testing before delivering.
- Eager to learn quickly and adapt to new tools, technologies, and domains — whether that is systems programming, data analysis, embedded systems, or application development.